

Chapter 4

Titanium, Magnesium, Nickel and Copper



Figure: A350 XWB Fuselage courtesy of Airbus

Titanium, Copper, Nickel and Magnesium

- I. Titanium
- II. Copper
- III. Nickel
- IV. Magnesium
- V. Ceramics

Students will be able to:

1. List the composition of the metals and their alloys
2. List the classification and uses of the metals and their alloys
3. Outline the properties and uses for different types of metals and their alloys
4. Describe the heat treatment available to apply on titanium and its alloys
5. Outline the properties and uses for ceramics

I. Titanium

- An English priest named Gregot discovered titanium.
- A crude separation of titanium ore was accomplished in 1825.
- In 1932, an extraction process was developed and became the first commercial method for producing titanium.
- The United States Bureau of Mines began making titanium sponge in 1946, and 4 years later the melting process began.
- The use of titanium is widespread.
- It is used in many commercial enterprises and is in constant demand for such items as pumps, screens, and other tools and fixtures where corrosion attack is prevalent.
- In aircraft construction and repair, titanium is used for fuselage skins, engine shrouds, firewalls, longerons, frames, fittings, air ducts, and fasteners.

- Titanium is used for making compressor disks, spacer rings, compressor blades and vanes, through bolts, turbine housings and liners, and miscellaneous hardware for turbine engines.
- Titanium, in appearance, is like stainless steel.
- One quick method used to identify titanium is the spark test.
- Titanium gives off a brilliant white trace ending in a brilliant white burst.
- Also, moistening the titanium and using it to draw a line on a piece of glass can accomplish identification. This leaves a dark line similar in appearance to a pencil mark.

- Titanium falls between aluminum and stainless steel in terms of elasticity, density, and elevated temperature strength.
- It has a melting point from 2,730 °F to 3,155 °F, **low thermal conductivity**, and a **low coefficient of expansion**.
- It is **light, strong (exceptional strength to weight ratio)**, and resistant to stress corrosion cracking.
- Titanium has **relatively low density**: 4500 kg/m³, approximately 60 percent heavier than aluminum and about 50 percent lighter than stainless steel.
- Because of the high melting point of titanium, high temperature properties are disappointing. The ultimate yield strength of titanium drops rapidly above 800 °F.
- It has **excellent corrosion resistance** and **high specific strength (strength/density ratio)**.

- The absorption of oxygen and nitrogen from the air at temperatures above 1,000 °F makes the metal so brittle on long exposure that it soon becomes worthless.
- However, titanium does have some merit for short time exposure up to 3,000 °F where strength is not important. Aircraft firewalls demand this requirement.
- Titanium is **nonmagnetic** and has an electrical resistance comparable to that of stainless steel. Some of the base alloys of titanium are quite hard.
- Heat treating and alloying do not develop the hardness of titanium to the high levels of some of the heat-treated alloys of steel.
- Titanium when compared to Aluminium or Steel or their alloys are considered to be **more expensive per weight**.

- It was only recently that a heat-treatable titanium alloy was developed. Prior to the development of this alloy, heating and rolling was the only method of forming that could be accomplished.
- However, it is possible to form the new alloy in the soft condition and heat-treat it for hardness.
- Iron, molybdenum, and chromium are used to stabilize titanium and produce alloys that quench-harden and age harden.
- The addition of these metals also adds ductility.
- The fatigue resistance of titanium is greater than that of aluminium or steel.

- Titanium becomes **softer as the degree of purity is increased.**
- It is not practical to distinguish between the various grades of commercially pure or unalloyed titanium by chemical analysis; therefore, the grades are determined by mechanical properties.

Titanium Designations

- The **A-B-C** classification of titanium alloys was established to provide a convenient and simple means of describing all titanium alloys.
- Titanium and titanium alloys possess three basic types of crystals: **A (alpha)**, **B (beta)**, and **C (combined alpha and beta)**.
- Their characteristics are:
 - **A (alpha)**—all-around performance; good weld ability; tough and strong both cold and hot; and resistant to oxidation.
 - **B (beta)**—bendability; excellent bend ductility; strong both cold and hot, but vulnerable to contamination.
 - **C (combined alpha and beta for compromise performances)**—strong when cold and warm, but weak when hot; good bendability; moderate contamination resistance; excellent forge ability.

- Titanium is manufactured for commercial use in two basic compositions: commercially-pure titanium and alloyed titanium.
- A-55 is an example of commercially-pure titanium.
- It has yield strength of 55,000 to 80,000 psi and is a general purpose grade for moderate to severe forming.
- It is sometimes used for nonstructural aircraft parts and for all types of corrosion-resistant applications, such as tubing.
- Type A-70 titanium is closely related to type A-55 but has yield strength of 70,000 to 95,000 psi. It is used where higher strength is required, and it is specified for many moderately stressed aircraft parts.
- For many corrosion applications, it is used interchangeably with type A-55. Both type A-55 and type A-70 is weldable.

Corrosion Characteristics

- The corrosion resistance of titanium deserves special mention. The resistance of the metal to corrosion is caused by the formation of a protective surface film of stable oxide or chemi-absorbed oxygen.
- Film is often produced by the presence of oxygen and oxidizing agents.
- **Corrosion of titanium is uniform.**
- There is little evidence of pitting or other serious forms of localized attack. Normally, it is not subject to stress corrosion, corrosion fatigue, intergranular corrosion, or galvanic corrosion.
- Its **corrosion resistance is equal or superior to 18-8 stainless steel.**

- An important characteristic of titanium is the reversible transformation of the crystal structure from alpha (α , hexagonal close-packed) structure to beta (β , body-centered cubic) structure when the temperatures exceed certain level.
- This allotropic behavior, which depends on the type and amount of alloy contents, allows complex variations in microstructure and more diverse strengthening opportunities than those of other nonferrous alloys such as copper or aluminum.
- Because the allotropic behavior of titanium allows diverse changes in microstructures by variations in thermo-mechanical processing, a broad range of properties and applications can be developed. This is especially true of the alloys with a two-phase, $\alpha + \beta$, crystal structure.

Effect of Alloying Elements

- One of the primary effects of alloying elements used in titanium production is the affect on the alpha to beta transformation temperature.
- Some elements raise the alpha to beta transformation temperature thereby stabilising the alpha crystal structure.
- While other elements lower the alpha to beta transformation temperature thereby stabilising the beta crystal structure.

- Titanium alloys can be classified per below:-
 - α -Ti Alloys
 - Near- α Ti Alloys
 - α - β Ti Alloys
 - β -Ti Alloys
- **α -Stabilizing Elements:**
 - Aluminium (Al), Tin (Sn), Zirconium (Zr), Oxygen (O) and Nitrogen (N)
- **β -Stabilizing Elements:**
 - Vanadium (V), Molybdenum (Mo), Silicon (Si) and Chromium (Cr)

- For Alpha (α) Stabilisers, such as Al, Sn, Zr, O, N and C, it is usual to express their alpha (α) stabilising effect as:
 - Aluminium (Al) Equivalent = $\text{Al} + (1/3) \text{Sn} + (1/6) \text{Zr} + 10 (\text{O} + 2\text{N} + \text{C})$
 - Note that zirconium (Zr) forms a solid solution with titanium and therefore makes the titanium alloys stronger.
- For Beta (β) Stabilisers, such as Mo, V, Nb, Ta, W, Fe, Cr, Mn, Ni and Co, it is usual to express their beta (β) stabilising effect as:
 - Molybdenum (Mo) Equivalent = $\text{Mo} + (0.67) \text{V} + (0.28) \text{Nb} + (0.2) \text{Ta} + (0.4) \text{W} + (1.25) \text{Cr} + (1.7) \text{Mn} + (1.25) \text{Ni} + (1.7) \text{Co}$

I. Titanium

Titanium and Titanium Alloys

Guidelines for Titanium Alloy Classification			
Class	Al-equiv	Mo-equiv	Properties/Applications
Alpha	< 8	< 1	Low Strength and corrosion resistant
Near-Alpha	8-10	< 2	High Strength at elevated temperature
Alpha/Beta	5-10	2-8	General purpose, high strength and good toughness
Near-Beta	< 8	10-15	High strength, high toughness and good forgeability
Metastable Beta	< 6	15-30	Very high strengths up to 1800 MPa

α -Ti Alloys:

- Example: Ti-5Al-2.5Sn
- Strong and they maintain their strength at high temperature (550 °C)
- Difficult to work/fabricate as they tend to form brittle intermetallic compound (Ti₃Al)
- Good weldability

Near- α Ti Alloys:

- Example: Ti-8Al-1Mo-1V
- Entirely α -phase with a small amount of β -phase dispersed throughout and they are normally used in the annealed condition.
- High strength, good creep resistance and toughness up to 850 °C.
- Good weldability.
- Airframe and jet engine parts.
- Aging after solution heat treatment increases the strength but makes alloy susceptible to stress-corrosion cracking in salt water environments.

α - β Ti Alloys

- Example: Ti-6Al-4V
- Aluminium stabilizes the α -phase vanadium stabilizes the β -phase
- Such alloys are solution treated, quenched and aged for increased strength
- Used for both high and low temperature applications → rocket motor cases, turbine blades and cryogenic vessels.

β -Ti Alloys

- Example: Ti-13V-11Cr-3Al
- The entire structure of such alloys consists of β -phase at room temperature after quenching (& air cooling)
- They can be cold-worked (unlike α -alloys) in the above heat-treated condition and aged to give very high strength with low ductility
- They also suffer from poor fatigue resistance and thus not widely used as the α - β alloys (the latter show better fatigue resistance).
- Aerospace components, honeycomb panels and high strength fasteners.

Limitations / Disadvantages

- One major problem with titanium is the **difficulty of processing it into usable shapes**. This is because many processing techniques involve elevated temperatures.
- **At high temperatures**, gases in the air start to dissolve into titanium alloys. This makes the alloys **brittle** and leaves **an oxide scale on its surface**.
- To avoid such problems, titanium alloys need to be processed (for example, annealed) **under high order of vacuum**. This of course increases its manufacturing costs.

Welding Behaviour of Titanium Alloys

- Welding has the greatest potential for affecting material properties.
- In all types of welds, contamination by interstitial impurities such as oxygen and nitrogen must be minimized to maintain useful ductility in the weldment.
- Alloy composition, welding procedure, and subsequent heat treatment are highly important in determining the final properties of welded joints.

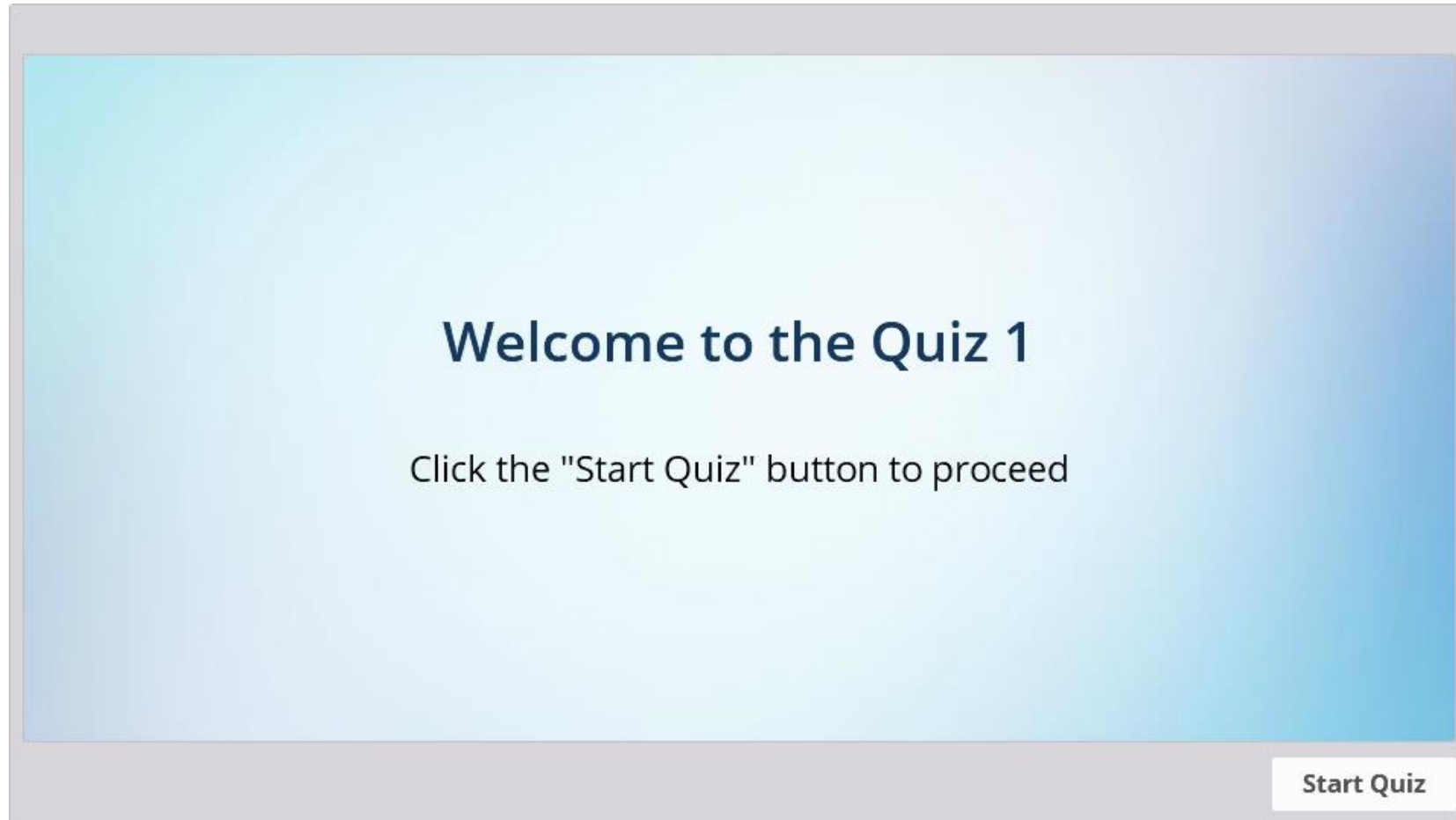
- Titanium and titanium alloys are **heat-treated** for the following purposes:
 - To **reduce residual stresses** developed during fabrication.
 - To **produce an optimal combination** of ductility, machinability, and dimensional and structural stability (annealing).
 - To **increase strength** (solution treating and aging)
 - To **optimise special properties** such as fracture toughness, fatigue strength, and high-temperature creep strength.

Heat-Treatment

- Various types of annealing treatments (single, duplex, beta, and recrystallization annealing, for example), and solution-treating and aging treatments, are used to achieve selected mechanical properties.
- **Stress relieving** and **annealing** may be employed to:
 - prevent preferential chemical attack in some corrosive environments;
 - prevent distortion (a stabilization treatment);
 - condition the metal for subsequent forming and fabricating operations.
- The response of titanium and titanium alloys to heat treatment depends on the composition of the metal and the effects of alloying elements on the α - β crystal transformation of titanium.
- In addition, **not all heat-treating cycles are applicable to all titanium alloys**, because the various alloys are designed for different purposes.

Quiz

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II. Copper

- Copper is one of the most widely distributed metals.
- It is the **only reddish-colored metal** and is **second only to silver in electrical conductivity**.
- Its use as a structural material is limited because of its great weight. However, some of its outstanding characteristics, such as its high electrical and heat conductivity, in many cases overbalance the weight factor.
- Copper is **very malleable** and **ductile (ease of fabrication)**, thus copper is ideal for making wire.
- Copper has **good corrosion resistance** although it **can be corroded by salt water**.
- For cast copper, the tensile strength is about 25,000 psi, and when cold rolled or cold drawn, its tensile strength increases to a range of 40,000 to 67,000 psi.

Unalloyed Copper (Pure Copper)

- Oxygen-Free High Conductivity (OFHC) Copper → O₂ content is very low (less than 10ppm) and that will not make copper brittle at elevated temperature
- The purest copper (99.92% min.)
- Known as a highest conductivity copper
- Used in electrical wire/equipment, radiators, refrigeration coils and heat exchangers

Unalloyed Copper (Pure Copper)

- Tough Pitch Copper → Contains a small amount of oxygen (0.04% O₂)
- Can withstand the ravages of time and weather because it does not harden with age and develop cracks
- Used for electrical wire/equipment, roofing, auto radiators and decorative metalwork

Unalloyed Copper (Pure Copper)

- Deoxidised Copper → Also known as phosphorous deoxidised copper
- By adding small amount of 0.04% phosphorus, oxide is removed
- Electrical conductivity is reduced by 25% as a result
- Used for domestic and industrial gas and water piping

II. Copper

Copper and Copper Alloys

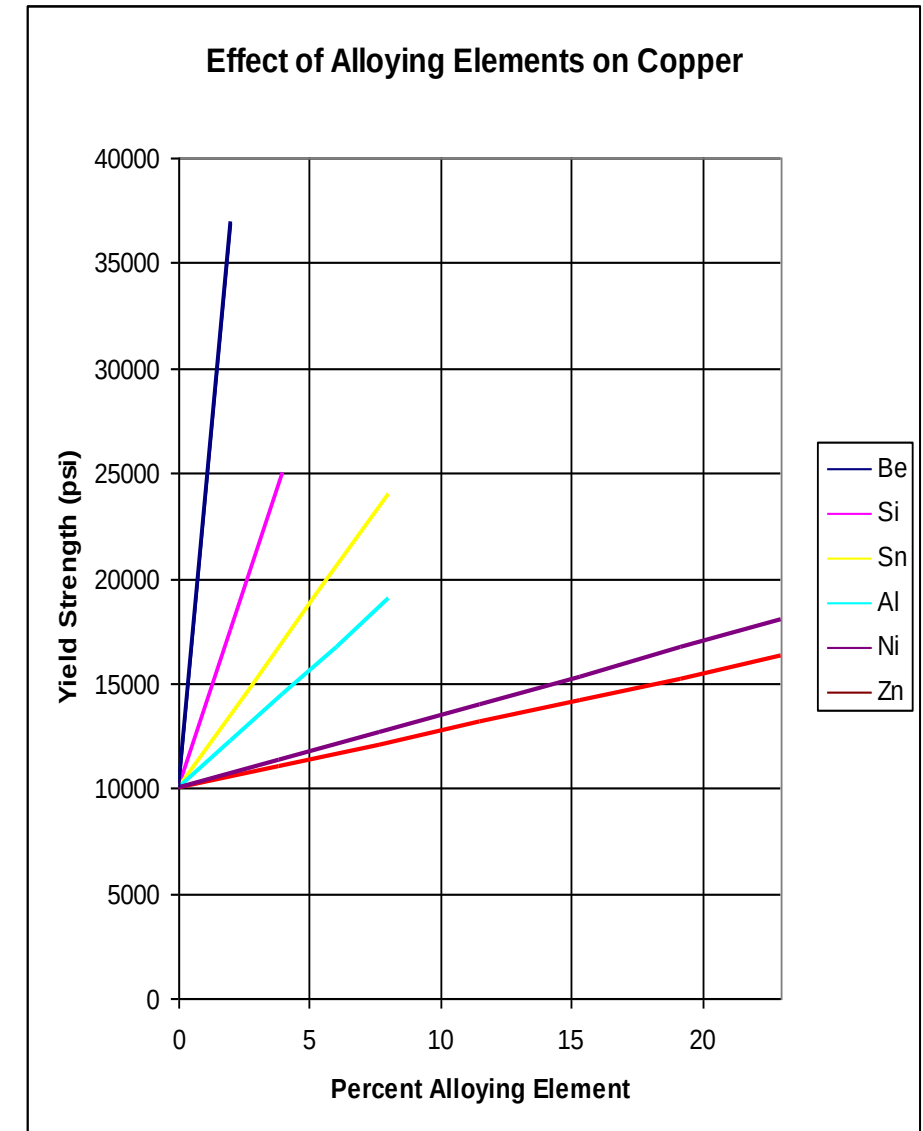
Effect Of Adding An Impurity

Tensile strength → Increases

Yield Strength → Increases

% Elongation → Decreases

Electrical Conductivity → Decreases



II. Copper

Copper and Copper Alloys

- In aircraft, **copper is used primarily in the electrical system** for bus bars, bonding, and as lock wire.
- **Beryllium copper (Cu-Be)** is one of the most successful of all the copper base alloys. It is a recently developed alloy containing about 97 percent copper, 2 percent beryllium, and sufficient nickel to increase the percentage of elongation.
- Cu-Be alloys obtain strength from precipitation hardening and can be heat-treated and cold-worked to produce very high tensile strength.
- Cu-Be alloys are used for tools requiring **high hardness, good corrosion resistance and strength**.
- The **resistance** of beryllium copper to **fatigue** and **wear** makes it suitable for diaphragms, precision bearings and bushings, ball cages, and spring washers.

- **Brass** is a **copper alloy containing zinc** and small amounts of aluminum, iron, lead, manganese, magnesium, nickel, phosphorous, and tin.
- Brass with a **zinc content of 30 to 35 percent is very ductile**, but that containing 45 percent has relatively high strength.

Cartridge Brass

- **70%Cu - 30%Zn alloy**
- **Most ductile** of the copper-zinc alloy.
- **Ease of fabrication**
- It is used in sheet metal pressing, deep drawing, locks, ammunition shells, nails

Standard Brass (Yellow Brass)

- **65%Cu – 35%Zn alloy**
- **Moderately ductility**
- **Ease of fabrication**
- Used for sheet metal pressing, plumbing accessories, lamp fixtures

Muntz metal

- A brass composed of **60 percent copper and 40 percent zinc**.
- It has **excellent corrosion-resistant qualities in salt water**.
- Its strength can be **increased by heat treatment**.
- **Relatively harder** and **brittle**
- As cast, this metal has an ultimate tensile strength of 50,000 psi, and it can be elongated 18 percent.
- It is used in making bolts and nuts, as well as parts that come in contact with salt water.

- **Red brass**, sometimes termed “**bronze**” because of its tin content, is used in fuel and oil line fittings.
- **Bronzes are copper alloys containing tin**. The true bronzes have up to 25 percent tin, but those with less than 11 percent are most useful, especially for such items as tube fittings in aircraft.
- This metal has good casting and finishing properties and machines freely.

Tin Bronzes

- 94%Cu, 5-10%Sn, 0.1-0.5%P alloys
- Also known as Phosphor Bronzes as up to 0.5% of phosphorous is added to deoxidize the metal
- **High strength, high corrosion resistance, low friction coefficient**
- Used for turbine blades, valve spindles, bearing bushes

Aluminium Bronzes

- Among the copper alloys are the copper aluminum alloys, of which the aluminum bronzes rank very high in aircraft usage.
- They would find greater usefulness in structures if it were not for their strength-to-weight ratio as compared with alloy steels.
- Wrought aluminum bronzes are **almost as strong and ductile as medium carbon steel**, and they possess a **high degree of resistance to corrosion by air, salt water, and chemicals**.
- They are readily forged, hot or cold rolled, and many react to heat treatment.

Aluminium Bronzes

- These copper base alloys contain up to 16 percent of aluminum (usually 5 to 11 percent), to which other metals, such as iron, nickel, or manganese, may be added.
- Aluminum bronzes have **good tearing qualities, great strength, hardness, and resistance to both shock and fatigue.**
- Because of these properties, they are used for diaphragms, gears, and pumps.
- Aluminum bronzes are available in rods, bars, plates, sheets, strips, and forgings.

Aluminium Bronzes

- Cast aluminum bronzes, using about 89 percent copper, 9 percent aluminum, and 2 percent of other elements, have high strength combined with ductility and are resistant to corrosion, shock, and fatigue.
- Because of these properties, **cast** aluminum bronze is used in bearings and pump parts. These alloys are useful in areas exposed to salt water and corrosive gases.

Manganese Bronzes

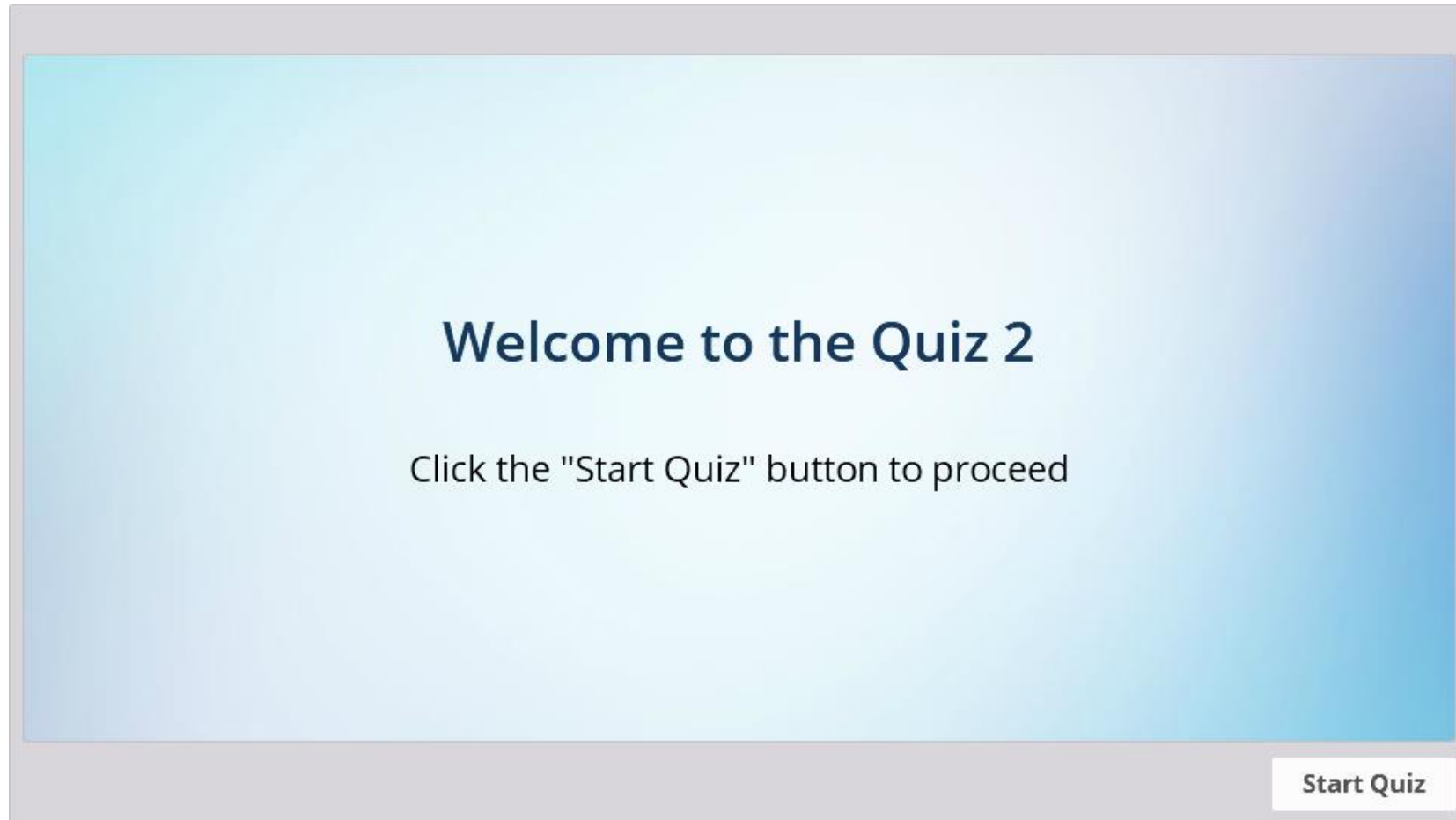
- Manganese bronze is an exceptionally **high strength, tough, corrosion-resistant** copper zinc alloy containing aluminum, manganese, iron, and occasionally, nickel or tin.
- This metal can be formed, extruded, drawn, or rolled to any desired shape. In rod form, it is generally used for machined parts for aircraft landing gears and brackets.

Silicon Bronzes

- Silicon bronze is a more recent development composed of about 95 percent copper, 3 percent silicon, and 2 percent manganese, zinc, iron, tin, and aluminum.
- Although not a bronze in the true sense because of its small tin content, silicon bronze has **high strength** and **great corrosion resistance**.

Quiz

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III. Nickel

- There are basically two nickel alloys used in aircraft: **Monel** and **Inconel**. Monel contains about 68 percent nickel and 29 percent copper, plus small amounts of iron and manganese.

Monel

- Monel, the leading high nickel alloy, combines the properties of **high strength** and **excellent corrosion resistance**. This metal consists of 68 percent nickel, 29 percent copper, 0.2 percent iron, 1 percent manganese, and 1.8 percent of other elements. It cannot be hardened by heat treatment.
- Nickel alloys can be **welded** or **easily machined**.
- Some of the nickel Monel, especially the nickel Monels containing small amounts of aluminum, are **heat-treatable to similar tensile strengths of steel**.

Monel

- When forged and annealed, it has a tensile strength of 80,000 psi. This can be increased by cold working to 125,000 psi, sufficient for classification among the tough alloys.
- Monel has been successfully used for gears and chains to operate retractable landing gears and for structural parts **subject to corrosion**.
- In aircraft, Monel is used for parts demanding both **strength** and **high resistance to corrosion**, such as exhaust manifolds and carburetor needle valves and sleeves.
- Nickel Monel is used in gears and parts that require **high strength** and **toughness**, such as exhaust systems that require high strength and corrosion resistance at elevated temperatures.

K-Monel

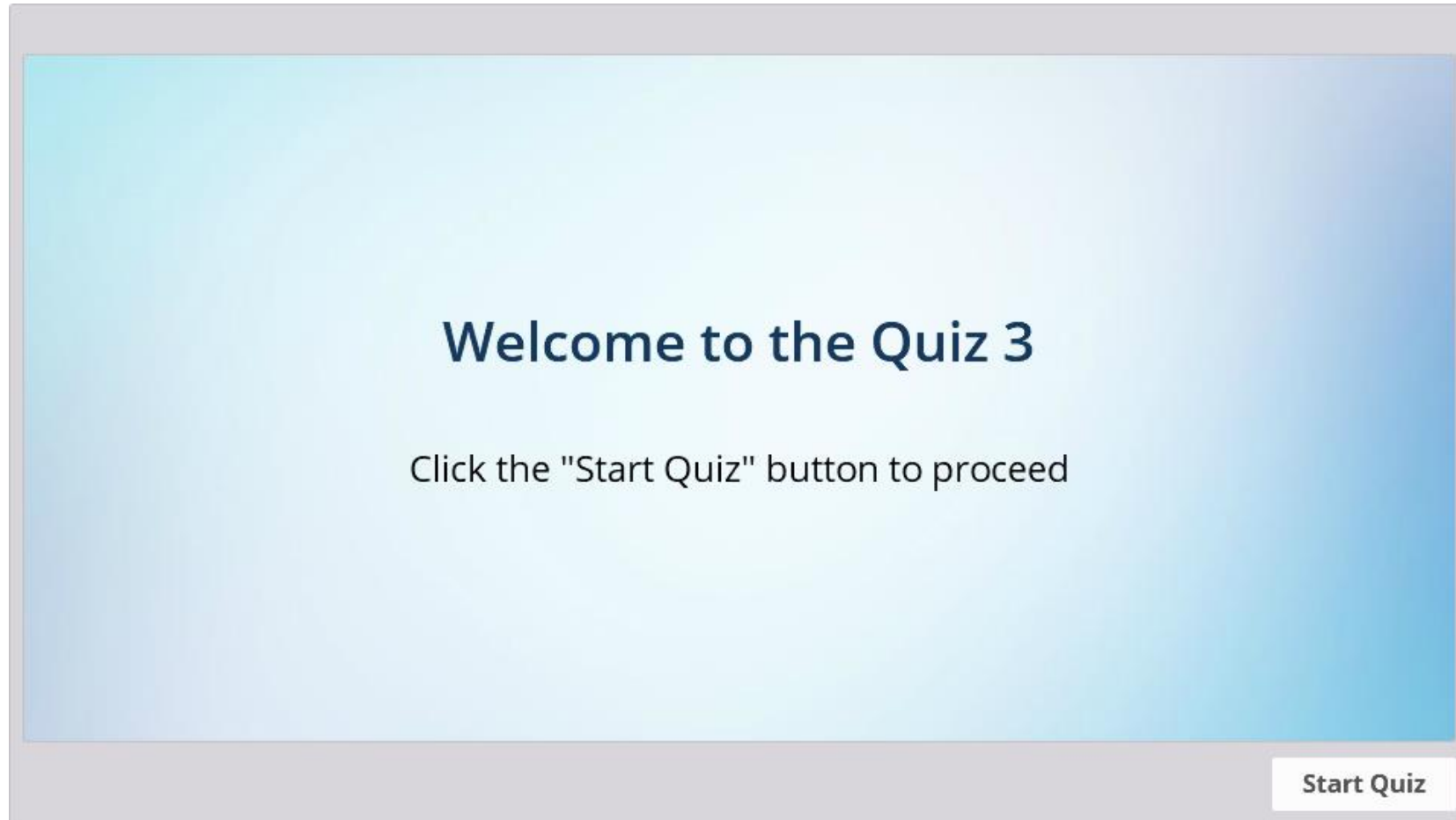
- K-Monel is a nonferrous alloy containing mainly nickel, copper, and aluminum. **Adding a small amount of aluminum to the Monel** formula produces it.
- It is **corrosion resistant** and **capable of being hardened by heat treatment**.
- K-Monel has been successfully used for gears and structural members in aircraft, which are subjected to corrosive attacks.
- This alloy is **nonmagnetic at all temperatures**.
- Both oxyacetylene and electric arc welding have successfully welded K-Monel sheet.

Inconel

- Inconel alloys of nickel produce a **high strength, high temperature** alloy containing approximately 80 percent nickel, 14 percent chromium, and small amounts of iron and other elements.
- The nickel Inconel alloys are frequently used in turbine engines because of their **ability to maintain their strength** and **corrosion resistance under extremely high temperature conditions**.
- Inconel and stainless steel are similar in appearance and are frequently found in the same areas of the engine.
- A common test is to apply one drop of cupric chloride and hydrochloric acid solution to the unknown metal and allow it to remain for 2 minutes. At the end of the soak period, a shiny spot indicates the material is nickel Inconel, and a copper-colored spot indicates stainless steel.

Quiz

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IV. Magnesium

- **Magnesium**, the world's lightest structural metal, is a silvery white material weighing **only two-thirds as much as aluminum**.
- Magnesium does not possess sufficient strength in its pure state for structural uses, but when alloyed with zinc, aluminum, and manganese, it produces an alloy having the **highest strength to-weight ratio** of any of the commonly used metals.
- Magnesium alloys also have **good high temperature mechanical properties** and **excellent corrosion resistance**.
- Magnesium is probably more widely distributed in nature than any other metal.

- Some of today's aircraft require more than one-half ton of this metal for use in hundreds of vital spots.
- Some wing panels are fabricated entirely from magnesium alloys, weigh 18 percent less than standard aluminum panels, and have flown hundreds of satisfactory hours.
- Among the aircraft parts that have been made from magnesium with a substantial savings in weight are nose wheel doors, flap cover skin, aileron cover skin, oil tanks, floorings, fuselage parts, wingtips, engine nacelles, instrument panels, radio masts, hydraulic fluid tanks, oxygen bottle cases, ducts, and seats.

- Magnesium alloys possess **good casting characteristics**. Their properties compare favorably with those of cast aluminum.
- In forging, hydraulic presses are ordinarily used, although, under certain conditions, forging can be accomplished in mechanical presses or with drop hammers.
- Magnesium alloys are **subject to such treatments** as annealing, quenching, solution heat treatment, aging, and stabilizing.
- Sheet and plate magnesium are annealed at the rolling mill.
- The solution heat treatment is used to put as much of the alloying ingredients as possible into solid solution, which results in high tensile strength and maximum ductility.
- Aging is applied to castings following heat treatment where maximum hardness and yield strength are desired.

IV. Magnesium

Magnesium and Magnesium Alloy

Designation of Magnesium Alloys according to ASTM B 275

First Part	Second Part	Third Part	Fourth Part
Indicates the two principal alloying elements	Indicates the amounts of the two principal alloying elements	Distinguishes between different alloys with the same percentages of the two principal alloying elements	Indicates condition (temper)
Consists of two code letters representing the two main alloying elements arranged in order of decreasing percentage (or alphabetically if percentages are equal)	Consists of two numbers corresponding to rounded-off percentages of the two main alloying elements and arranged in same order as alloy designations in first part	Consists of a letter of the alphabet assigned in order as compositions become standard	Consists of a letter followed by a number (separated from the third part of the designation by a hyphen)
A-aluminum B-bismuth C-copper D-cadmium E-rare earth F-iron G-magnesium H-thorium K-zirconium L-lithium M-manganese N-nickel P-lead Q-silver R-chromium S-silicon T-tin W-yttrium Y-antimony Z-zinc	Whole numbers	Letters of alphabet except I and O	F-as fabricated O-as annealed H10 and H11- slightly strain hardened H23,H24 and H26- strain hardened and partially annealed T4-solution heat treated T5-artificially aged only T6-solution heat treated and artificially aged T8-solution heat treated, cold worked and artificially

As an example of this designation system, consider magnesium alloy **AZ81A-T4**.

- The first part of the designation, **AZ**, signifies that aluminium and zinc are the two principal alloying elements.
- The second part of the designation, **81**, gives the rounded-off percentages of aluminium and zinc (**8** and **1**, respectively).
- The third part, **A**, indicates that it is the first alloy standardized with **8% Al** and **1% Zn** as the principal alloying additions.
- The fourth part, **T4**, denotes that the alloy is solution heat-treated.

V. Ceramics

- The term ceramics means dishes, porcelain vases to most.
- **Ceramics are defined to be solids composed of compounds which contain metallic and non-metallic elements held together with strong atomic forces. (ionic or covalent bonds).**
- Bricks and clay products are example of ceramics:-
 - Mostly composed of silica, alumina and water (Al_2O_3 , H_2O , 4SiO_2)
 - The bonding mechanism changes from weak van der Waals force to strong ionic or covalent bonds after firing.
- Uses of ceramics started off in the early 1980s and established in the 2000s.
- Most engineering ceramics are not clay products but mostly oxides, nitrides and carbides which are sinistered to high density.

- Ceramics are known for the following properties:-
 - **High hardness** and **brittleness** (glass-like) due to strong ionic and covalent bonds
 - **Good refractoriness** (i.e. ability to withstand high temperature without deterioration in properties)
 - **Good thermal and electrical insulator**
 - **Good resistance to corrosion**
 - **Great stiffness greater than those of steel**



Powder



Slurry

<https://sciengineeredmaterials.com/ceramic-powders/>



<https://global.kyocera.com/prdct/fc/product/category/metalize/metalize003.html>



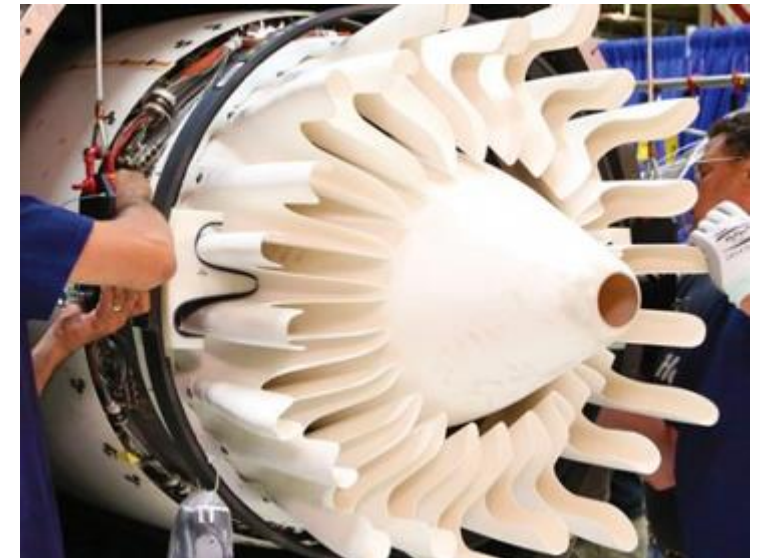
Porcelain

https://www.123rf.com/photo_101258486_colorful-ceramics-exposed-outside-of-a-souvenirs-shops.html



<http://ceramlet.com/>

Alumina



<https://advancedceramics.org/>

SiC compounds

The End

You haven't seen a tree until you've seen its shadow from the sky.

- Amelia Earhart